



PDF Download
3746252.3761484.pdf
18 January 2026
Total Citations: 0
Total Downloads: 111

 Latest updates: <https://dl.acm.org/doi/10.1145/3746252.3761484>

SHORT-PAPER

RerankArena: A Unified Platform for Evaluating Retrieval, Reranking and RAG with Human and LLM Feedback

ABDELRAHMAN ABDALLAH, University of Innsbruck, Innsbruck, Tyrol, Austria

MAHMOUD ABDALLA, Chungbuk National University, Cheongju, Chungcheongbuk-do, South Korea

BHAWNA PIRYANI, University of Innsbruck, Innsbruck, Tyrol, Austria

JAMSHID MOZAFARI, University of Innsbruck, Innsbruck, Tyrol, Austria

MOHAMMED ALI, University of Innsbruck, Innsbruck, Tyrol, Austria

ADAM JATOWT, University of Innsbruck, Innsbruck, Tyrol, Austria

Open Access Support provided by:

University of Innsbruck

Chungbuk National University

Published: 10 November 2025

[Citation in BibTeX format](#)

CIKM '25: The 34th ACM International Conference on Information and Knowledge Management
November 10 - 14, 2025
Seoul, Republic of Korea

Conference Sponsors:

SIGWEB
SIGIR

RankArena: A Unified Platform for Evaluating Retrieval, Reranking and RAG with Human and LLM Feedback

Abdelrahman Abdallah
abdelrahman.abdallah@uibk.ac.at
University of Innsbruck
Innsbruck, Tyrol, Austria

Mahmoud Abdalla
Mahmoud.Abdalla@cbnu.ac.kr
Chungbuk National University
Cheongju-si, Cheongju, Republic of
Korea

Bhawna Piryani
bhawna.piryani@uibk.ac.at
University of Innsbruck
Innsbruck, Tyrol, Austria

Jamshid Mozafari
jamshid.mozafari@uibk.ac.at
University of Innsbruck
Innsbruck, Tyrol, Austria

Mohammed Ali
Mohammed.ali@uibk.ac.at
University of Innsbruck
Innsbruck, Tyrol, Austria

Adam Jatowt
adam.jatowt@uibk.ac.at
University of Innsbruck
Innsbruck, Tyrol, Austria

Abstract

Evaluating the quality of retrieval-augmented generation (RAG) and document reranking systems remains challenging due to the lack of scalable, user-centric, and multi-perspective evaluation tools. We introduce RankArena, a unified platform for comparing and analysing the performance of retrieval pipelines, rerankers, and RAG systems using structured human and LLM-based feedback as well as for collecting such feedback. RankArena supports multiple evaluation modes: direct reranking visualisation, blind pairwise comparisons with human or LLM voting, supervised manual document annotation, and end-to-end RAG answer quality assessment. It captures fine-grained relevance feedback through both pairwise preferences and full-list annotations, along with auxiliary metadata such as movement metrics, annotation time, and quality ratings. The platform also integrates LLM-as-a-judge evaluation, enabling comparison between model-generated rankings and human ground truth annotations. All interactions are stored as structured evaluation datasets that can be used to train rerankers, reward models, judgment agents, or retrieval strategy selectors. Our platform is publicly available at <https://rankarena.ngrok.io/>, and the Demo video is provided¹.

CCS Concepts

• Information systems → Language models; Top-k retrieval in databases.

Keywords

Ranking, RAG, LLM-as-a-Judge, Retriever

ACM Reference Format:

Abdelrahman Abdallah, Mahmoud Abdalla, Bhawna Piryani, Jamshid Mozafari, Mohammed Ali, and Adam Jatowt. 2025. RankArena: A Unified Platform for Evaluating Retrieval, Reranking and RAG with Human and LLM Feedback. In *Proceedings of the 34th ACM International Conference on Information*

and Knowledge Management (CIKM '25), November 10–14, 2025, Seoul, Republic of Korea. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3746252.3761484>

1 Introduction

Recent advancements in large language models (LLMs) and retrieval-augmented generation (RAG) [24, 15] systems have significantly expanded the scope of natural language understanding and generation. These systems combine information retrieval with powerful generative models, enabling applications such as open-domain question answering [18, 40, 39, 26, 16], summarization [13], and chat assistants [31]. However, as models and pipelines become more complex, evaluating their performance in alignment with human preferences remains a persistent challenge. Conventional benchmarks [37, 38] often relying on static datasets and predefined metrics fail to capture user-centric notions of quality such as *document relevance*, *answer usefulness*, or *faithfulness in generation*. To address this gap, we introduce **RankArena**, a unified, open-source evaluation and annotation platform that supports multi-faceted assessment of *retrieval*, *reranking*, and *generation quality*. Inspired by recent efforts such as MT-Bench [5] and Chatbot Arena [11], our platform extends the evaluation paradigm beyond model-centric metrics by integrating *human-in-the-loop* and *LLM-as-a-judge* feedback [42, 17]. Users interact with multiple reranking and RAG pipelines across various tasks, cast preferences in blind pairwise comparisons, annotate ranked document lists, and evaluate generated answers, all of which are structured into reusable datasets.

RankArena features five complementary evaluation modes:

- (1) **Reranker Comparison:** Direct and blind pairwise battles between ranked document lists.
- (2) **Manual Annotation:** Supervised full-list ranking with quality labels, tracking annotation effort and movement metrics.
- (3) **LLM Judgment:** GPT-based voting on document orderings or RAG answers.
- (4) **Comprehensive Reranker Leaderboard:** Evaluation and comparison of multiple rerankers across diverse retrieval tasks, using shared datasets and unified metrics.
- (5) **RAG Output Evaluation:** Qualitative and preference-based evaluation of generated answers from end-to-end RAG pipelines.

¹<https://youtu.be/jIYAP4PaSSI>



This work is licensed under a Creative Commons Attribution 4.0 International License. *CIKM '25, Seoul, Republic of Korea*

© 2025 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-XXXX-X/2018/06

<https://doi.org/10.1145/3746252.3761484>

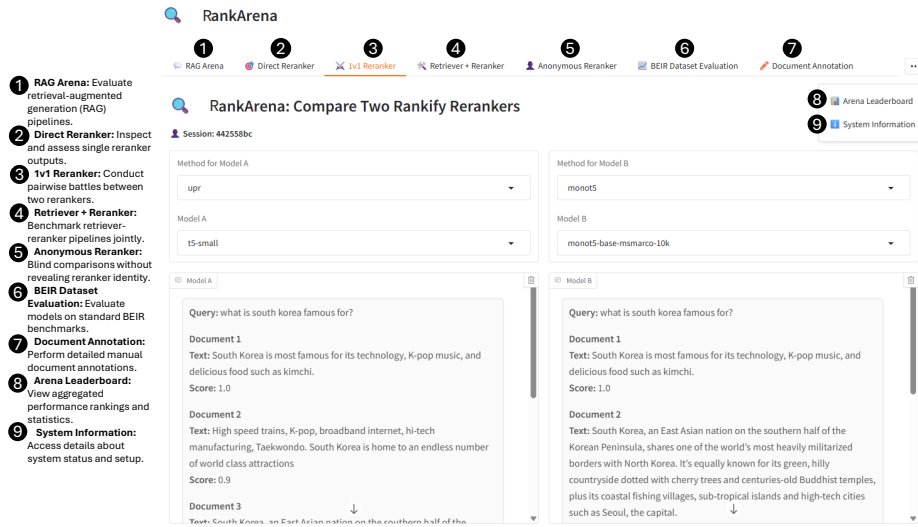


Figure 1: Screenshot of the RankArena interface with added explanations of each tab (1-9) showcasing the system’s main evaluation modes and tools.



Figure 2: The overview of RankArena highlighting its main evaluation modules.

- (6) **Dataset Collection for Reward and Judge Model Training:** Systematically collect aligned human and LLM preference data, creating reusable datasets to train reward models for retrieval and to build or fine-tune LLM judges capable of comparing retrieval or RAG systems effectively.

Unlike previous works that focus solely on chatbot alignment or answer preference [11, 25], RankArena supports *retrieval-specific evaluation signals*, making it uniquely suited to train and benchmark rerankers, reward models, and retrieval agents. Our platform captures diverse signals, including user preferences, annotation metadata, rank movements, generation ratings, and LLM feedback, all stored in a structured, extensible dataset format. These datasets can be used for *training new models*, analyzing *alignment gaps* between humans and LLMs, or developing *retrieval strategies adaptive to user feedback*.

2 RankArena Functionality

RankArena fills this gap by providing a unified platform for collecting pairwise preferences, listwise annotations, and LLM-as-a-judge feedback at scale. This system is important because it enables: (i) creation of reusable datasets for training supervised rerankers and reward models; (ii) study of human-LLM agreement in ranking tasks; (iii) adaptive benchmarking that evolves with real user queries and feedback; and (iv) reproducible, open evaluations combining human and automated signals.

The overview of our system is illustrated in Figure 1 and 2, which highlights the various modules working in concert to support rich evaluation scenarios. Our platform facilitates blind head-to-head comparisons, direct inspection of retrieval outputs, large-scale annotation, and statistical aggregation of pairwise preferences into global model rankings.

Arena Battles and Preference Collection: At the heart of RankArena lies the **1v1 Arena**, a head-to-head evaluation interface where two rerankers, retrieval systems, or RAG pipelines are presented in a blind and randomized manner. Either human or LLM-based judges interact with the system by comparing the outputs of these two systems on the same query and voting for the preferred result. This design simplifies the cognitive load on evaluators compared to absolute scoring, as it only requires relative preference judgments. We support over 24 reranking methods [32, 7, 20, 10, 23, 14, 22, 12, 30, 29, 36, 36, 43, 27, 2, 1] spanning pairwise, listwise, and pointwise approaches, with a total of 84 models available for evaluation in these head-to-head battles.

LLM-as-a-Judge Evaluation: To complement human annotations, RankArena integrates an **LLM judge** module, where large language models (e.g., GPT-4) provide automated pairwise preferences between retrieval or RAG outputs. The LLM-as-a-judge mechanism enables rapid scaling of evaluations and offers a cost-effective alternative to human voting. We adopt a structured prompting strategy

that asks LLMs to consider relevance, ranking order quality, and overall usefulness in answering the query. The prompt clearly instructs the LLM to review two reranked document lists for a given query and respond strictly within a predefined format, specifying the winning reranker and providing concise reasoning. This approach ensures consistency, interpretability, and comparability of LLM judgments across different evaluation tasks.

End-to-End RAG: RankArena also supports the evaluation of **full RAG pipelines**, where annotators or LLM judges assess generated answers along with their supporting document lists. This integrated evaluation allows for measuring not only document ranking quality but also how well retrieved evidence supports final answer generation. Such evaluation is essential for understanding system performance in complex, multi-stage pipelines. We support five different retrievers (DPR [21], Colbert [22], Contreiver [19], BGE [8], BM25 [33]) operating over two indexing corpora—**Wikipedia** and **MS MARCO** as well as online retrievers for live document retrieval, providing flexibility across static and dynamic knowledge sources.

For RAG evaluations, we extend our **LLM-as-a-Judge** module to assess both the generated answers and their associated supporting documents from two different rerankers. The LLM judge considers not only the relevance and order of retrieved contexts but also the quality of the final answers and their alignment with the retrieved evidence. This method enables scalable, automated assessment of how effectively different rerankers support answer generation in RAG systems. We adopt a structured prompt for RAG evaluations that asks the LLM judge to compare two RAG outputs each consisting of a document list, a generated answer, and an indication of which document the answer was primarily drawn from. The LLM is instructed to evaluate results based on: (1) Relevance of retrieved documents, (2) Quality of document ranking, (3) Usefulness and correctness of the generated answers, and (4) Faithfulness of the answers to the supporting documents. This structured prompting is illustrated in Figure 3, enabling consistent and interpretable LLM voting for end-to-end RAG pipelines.

Direct Reranker and Full-List Annotation: Beyond head-to-head comparisons, RankArena supports two complementary modes for evaluating single reranker outputs. In the **direct reranker** mode, users provide both the query and the candidate documents, and the system applies a single reranker to produce a ranked list. This mode allows users to directly examine how a specific reranker orders documents for a given query. In contrast, the **full-list annotation** mode involves users providing a query (or selecting one), after which the system retrieves documents from online or offline corpora. The user then manually reorders or assigns relevance grades to these documents, creating high-quality listwise supervision data. This annotated data can be used to train supervised rerankers, listwise ranking models, or reward models for preference learning.

Leaderboard Aggregation and Statistical Modeling: The preferences collected from head-to-head battles, direct evaluations, and LLM judges are aggregated into a **comprehensive reranker leaderboard**. We compute win rates $w_i = \frac{\text{wins}_i}{\text{total votes}_i}$ and transform these into an ELO-style rating:

$$R_i = 1200 + 32 \cdot (w_i - 0.5) \cdot \min(\log(\text{total votes}_i + 1), 5.0)$$

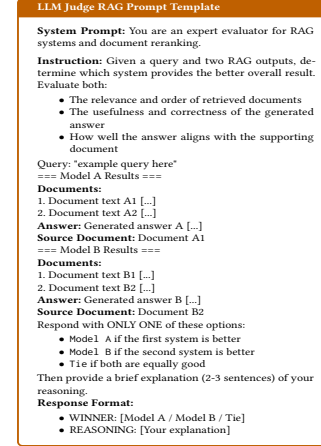


Figure 3: Illustration of our LLM Judge RAG prompt template. The LLM evaluates RAG results by comparing retrieved documents, generated answers, and evidence alignment.

This formulation reflects both win rate and the confidence in that estimate (via number of votes). Traditional benchmark scores, such as BEIR averages, are calculated as: $\text{BEIR Avg}_i = \frac{1}{N} \sum_{j=1}^N S_{ij}$ where S_{ij} is the score on benchmark j . Models are ranked by descending ELO rating, with BEIR performance as a secondary criterion. Our leaderboard thus integrates user and LLM preference signals with standardized benchmarks, offering a holistic view of model performance.

3 Results and Analysis

Experiment Setup. Our experiments were conducted using the **RankArena** platform, developed on top of the Rankify [3], Gradio [4] and PyTorch [28] framework. The system is deployed on a server equipped with two NVIDIA A40 GPUs and 250 GB of RAM, providing sufficient resources to support continuous 24-hour operation for large-scale evaluation and annotation.

RankArena integrates diverse reranking and retrieval components for comprehensive benchmarking. The system currently includes: (1) **24 reranking methods** spanning pointwise, pairwise, and listwise approaches, (2) **84 reranker models** (as some methods apply different models) evaluated across different retrieval tasks, and (3) **5 retrievers** operating on two corpora: **Wikipedia** and **MS MARCO**, along with online retrieval for dynamic evaluation. Below, we report the results and analyses based on the following **Arena Statistics**: (1) **Total User Votes**: 102. (2) **Active Models**: 36 (models that have received human votes). (3) **Benchmark Models**: 80 (models with BEIR benchmark scores). (4) **Methods Tested**: 25. (5) **Average User-LLM Agreement**: 74.2%

BEIR Benchmark Correlations. We analyze the correlation between different BEIR benchmark [37] datasets to understand how model performance generalizes across tasks. Figure 4 (available also in the leaderboard) illustrates the correlation matrix between BEIR Average and individual benchmarks (DL19, DL20, Covid, NFCorpus, Touche, DBPedia, SciFact).

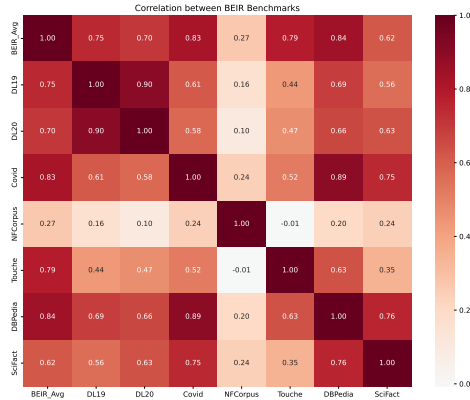


Figure 4: Correlation matrix between BEIR Average and individual benchmarks.

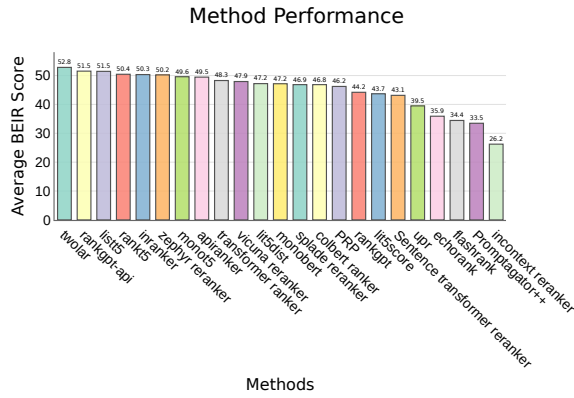


Figure 5: Average BEIR performance of different reranking methods.

The BEIR Average shows strong correlations with DBPedia (0.84), Covid (0.83), and DL19 (0.75), indicating these datasets contribute substantially to the aggregate BEIR metric. In contrast, NFCorpus displays weak correlation (0.27) with BEIR Average and other benchmarks, suggesting it measures distinct capabilities. Such correlation patterns provide insight into the diversity of tasks within BEIR and the challenges of designing universal rankers.

Models Performance on BEIR. Figure 5 (also displayed in the leaderboard tab) presents the average BEIR scores across evaluated methods. The highest-performing method, *two1ar* [6], achieves an average score of 52.8, while other top methods (e.g., *rankgpt-api* [35], *listt5* [41]) exhibit scores above 50. Methods like *UPR* [34] and *incontext reranker* [9] perform considerably worse, reflecting a gap between the strongest and weakest reranking strategies.

Human-LLM Agreement. We assess next the alignment between human votes and LLM judge decisions. Figure 6 shows a scatter plot of human preference rates versus LLM judge rates across the evaluated models. The results demonstrate a generally strong positive correlation between human and LLM judgments, with most models

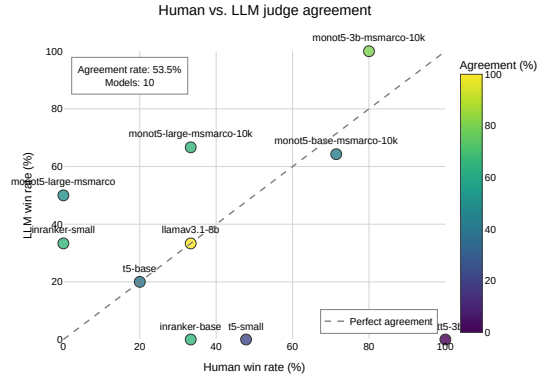


Figure 6: Scatter plot of human-LLM agreement across models.

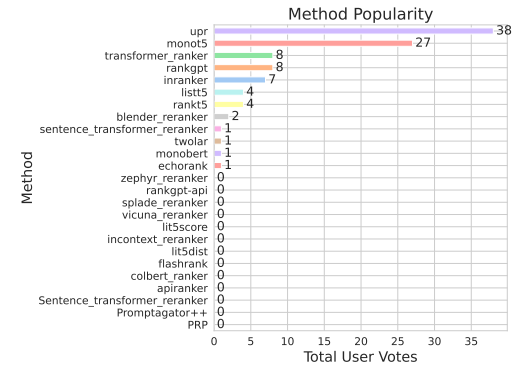


Figure 7: Distribution of user votes across different reranking methods in RankArena. The figure shows that a few methods receive

clustering near the diagonal line. Notable deviations are observed for certain models, indicating areas where human preferences and automated judgments diverge, warranting deeper qualitative analysis.

Method Popularity and User Engagement. Figure 7 (shown also under the leaderboard tab) illustrates the popularity of different reranking methods in RankArena, measured by the total number of user votes received during head-to-head evaluations. Our data shows a clear skew in user engagement across methods: a small number of rerankers dominate in terms of participation, while many others receive few or no votes.

4 Conclusion

We presented RankArena, a unified platform for evaluating retrieval, reranking, and RAG systems using human and LLM feedback. By combining pairwise preferences, full-list annotations, and automated judgments, RankArena enables the creation of reusable datasets and open benchmarking of retrieval pipelines.

GenAI Usage Disclosure

We used OpenAI's ChatGPT for minor language editing, specifically to rephrase sentences and correct grammatical errors.

References

- [1] Abdelrahman Abdallah, Jamshid Mozafari, Bhawna Pirani, Mohammed M Abdelgawad, and Adam Jatowt. 2024. Dynrank: improving passage retrieval with dynamic zero-shot prompting based on question classification. *arXiv preprint arXiv:2412.00600*.
- [2] Abdelrahman Abdallah, Jamshid Mozafari, Bhawna Pirani, and Adam Jatowt. 2025. Asrank: zero-shot re-ranking with answer scent for document retrieval. *arXiv preprint arXiv:2501.15245*.
- [3] Abdelrahman Abdallah, Bhawna Pirani, Jamshid Mozafari, Mohammed Ali, and Adam Jatowt. 2025. Rankify: a comprehensive python toolkit for retrieval, re-ranking, and retrieval-augmented generation. *arXiv preprint arXiv:2502.02464*.
- [4] Abubakar Abid, Ali Abdalla, Ali Abid, Dawood Khan, Abdulrahman Alfozan, and James Zou. 2019. Gradio: hassle-free sharing and testing of ml models in the wild. *arXiv preprint arXiv:1906.02569*.
- [5] Ge Bai et al. 2024. Mt-bench-101: a fine-grained benchmark for evaluating large language models in multi-turn dialogues. *arXiv preprint arXiv:2402.14762*.
- [6] Davide Baldelli, Junfeng Jiang, Akiko Aizawa, and Paolo Torroni. 2024. Twolar: a two-step llm-augmented distillation method for passage reranking. In *European Conference on Information Retrieval*. Springer, 470–485.
- [7] Parishad Behnam Ghader, Vaibhav Adlakha, Marius Mosbach, Dzmitry Bahdanau, Nicolas Chapados, and Siva Reddy. 2024. Llm2vec: large language models are secretly powerful text encoders. (2024). <https://arxiv.org/abs/2404.05961> [cs. CL].
- [8] Jianlv Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. 2024. Bge m3-embedding: multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. *arXiv preprint arXiv:2402.03216*.
- [9] Shijie Chen, Bernal Jiménez Gutiérrez, and Yu Su. 2024. Attention in large language models yields efficient zero-shot re-rankers. *arXiv preprint arXiv:2410.02642*.
- [10] Zijian Chen, Ronak Pradeep, and Jimmy Lin. 2024. An early first reproduction and improvements to single-token decoding for fast listwise reranking. (2024). <https://arxiv.org/abs/2411.05508> arXiv: 2411.05508 [cs. IR].
- [11] Wei-Lin Chiang et al. 2024. Chatbot arena: an open platform for evaluating llms by human preference. In *Forty-first International Conference on Machine Learning*.
- [12] [SW] Prithiviraj Damodaran, FlashRank, Lightest and Fastest 2nd Stage Reranker for search pipelines. Version 1.0.0, Dec. 2023. doi:10.5281/zenodo.10426927, URL: <https://github.com/PrithivirajDamodaran/FlashRank>.
- [13] Darren Edge et al. 2024. From local to global: a graph rag approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*.
- [14] Thibault Formal, Carlos Lassance, Benjamin Piwowarski, and Stéphane Clinchant. 2022. From distillation to hard negative sampling: making sparse neural ir models more effective. In *Proceedings of the 45th international ACM SIGIR conference on research and development in information retrieval*, 2353–2359.
- [15] Yunfan Gao et al. 2023. Retrieval-augmented generation for large language models: a survey. *arXiv preprint arXiv:2312.10997*, 2, 1.
- [16] Raphael Gruber, Abdelrahman Abdallah, Michael Färber, and Adam Jatowt. 2024. Complextempqa: a large-scale dataset for complex temporal question answering. *arXiv preprint arXiv:2406.04866*.
- [17] Jiawei Gu et al. 2024. A survey on llm-as-a-judge. *arXiv preprint arXiv:2411.15594*.
- [18] Rujun Han, Yuhao Zhang, Peng Qi, Yumo Xu, Jinyuan Wang, Lan Liu, William Yang Wang, Bonan Min, and Vittorio Castelli. 2024. Rag-qa arena: evaluating domain robustness for long-form retrieval augmented question answering. *arXiv preprint arXiv:2407.13998*.
- [19] Gautier Izacard, Mathilde Caron, Lucas Hosseini, Sebastian Riedel, Piotr Bojanowski, Armand Joulin, and Edouard Grave. 2021. Unsupervised dense information retrieval with contrastive learning. (2021). doi:10.48550/ARXIV.2112.09118.
- [20] Dongfu Jiang, Xiang Ren, and Bill Yuchen Lin. 2023. Llm-blender: ensembling large language models with pairwise ranking and generative fusion. (2023). <https://arxiv.org/abs/2306.02561> arXiv: 2306.02561 [cs. CL].
- [21] Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick SH Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense passage retrieval for open-domain question answering. In *EMNLP (1)*, 6769–6781.
- [22] Omar Khattab and Matei Zaharia. 2020. Colbert: efficient and effective passage search via contextualized late interaction over bert. In *Proceedings of the 43rd International ACM SIGIR conference on research and development in Information Retrieval*, 39–48.
- [23] Thiago Laitz, Konstantinos Papakostas, Roberto Lotufo, and Rodrigo Nogueira. 2024. Inranker: distilled rankers for zero-shot information retrieval. (2024). <https://arxiv.org/abs/2401.06910> arXiv: 2401.06910 [cs. IR].
- [24] Patrick Lewis et al. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems*, 33, 9459–9474.
- [25] Haipeng Luo, Qingfeng Sun, Can Xu, Pu Zhao, Qingwei Lin, Jian-Guang Lou, Shifeng Chen, Yansong Tang, and Weizhu Chen. 2024. Wizardarena: post-training large language models via simulated offline chatbot arena. *Advances in Neural Information Processing Systems*, 37, 111544–111570.
- [26] Jamshid Mozafari, Abdelrahman Abdallah, Bhawna Pirani, and Adam Jatowt. 2024. Exploring hint generation approaches in open-domain question answering. *arXiv preprint arXiv:2409.16096*.
- [27] Rodrigo Nogueira, Wei Yang, Kyunghyun Cho, and Jimmy Lin. 2019. Multi-stage document ranking with bert. (2019). <https://arxiv.org/abs/1910.14424> [cs. IR].
- [28] A Paszke. 2019. Pytorch: an imperative style, high-performance deep learning library. *arXiv preprint arXiv:1912.01703*.
- [29] Ronak Pradeep, Sahel Sharifmoghaddam, and Jimmy Lin. 2023. Rankvicuna: zero-shot listwise document reranking with open-source large language models. (2023). <https://arxiv.org/abs/2309.15088> arXiv: 2309.15088 [cs. IR].
- [30] Ronak Pradeep, Sahel Sharifmoghaddam, and Jimmy Lin. 2023. Rankzephyr: effective and robust zero-shot listwise reranking is a breeze! (2023). <https://arxiv.org/abs/2312.02724> arXiv: 2312.02724 [cs. IR].
- [31] Mahimai Raja, E Yuvarajan, et al. 2024. A rag-based medical assistant especially for infectious diseases. In *2024 International Conference on Inventive Computation Technologies (ICICT)*. IEEE, 1128–1133.
- [32] Muhammad Shihab Rashid, Jannat Ara Meem, Yue Dong, and Vagelis Hristidis. 2024. Ecorank: budget-constrained text re-ranking using large language models. (2024). <https://arxiv.org/abs/2402.10866> arXiv: 2402.10866 [cs. CL].
- [33] Stephen Robertson, Hugo Zaragoza, et al. 2009. The probabilistic relevance framework: bm25 and beyond. *Foundations and Trends® in Information Retrieval*, 3, 4, 333–389.
- [34] Devendra Singh Sachan, Mike Lewis, Mandar Joshi, Armen Aghajanyan, Wen-tau Yih, Joelle Pineau, and Luke Zettlemoyer. 2022. Improving passage retrieval with zero-shot question generation. <https://arxiv.org/abs/2204.07496>.
- [35] Weiwei Sun, Lingyong Yan, Xinyu Ma, Shuaiqiang Wang, Pengjie Ren, Zhumin Chen, Dawei Yin, and Zhaochun Ren. 2023. Is chatgpt good at search? investigating large language models as re-ranking agents. *arXiv preprint arXiv:2304.09542*.
- [36] Manveer Singh Tamber, Ronak Pradeep, and Jimmy Lin. 2023. Scaling down, lifting up: efficient zero-shot listwise reranking with seq2seq encoder-decoder models. (2023). <https://arxiv.org/abs/2312.16098> arXiv: 2312.16098 [cs. IR].
- [37] Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. 2021. Beir: a heterogeneous benchmark for zero-shot evaluation of information retrieval models. *arXiv preprint arXiv:2104.08663*.
- [38] Ellen M Voorhees. 2001. The trec question answering track. *Natural Language Engineering*, 7, 4, 361–378.
- [39] Jonas Wallat, Abdelrahman Abdallah, Adam Jatowt, and Avishek Anand. 2025. A study into investigating temporal robustness of llms. *arXiv preprint arXiv:2503.17073*.
- [40] Yi Yang, Wen-tau Yih, and Christopher Meek. 2015. Wikiqa: a challenge dataset for open-domain question answering. In *Proceedings of the 2015 conference on empirical methods in natural language processing*, 2013–2018.
- [41] Soyoung Yoon, Eumbi Choi, Jiyeon Kim, Hyeonju Yun, Yireun Kim, and Seungwon Hwang. 2024. List5: listwise reranking with fusion-in-decoder improves zero-shot retrieval. *arXiv preprint arXiv:2402.15838*.
- [42] Lianmin Zheng et al. 2023. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in Neural Information Processing Systems*, 36, 46595–46623.
- [43] Honglei Zhuang, Zhen Qin, Rolf Jagerman, Kai Hui, Ji Ma, Jing Lu, Jianmo Ni, Xuanhui Wang, and Michael Bendersky. 2022. Rankt5: fine-tuning t5 for text ranking with ranking losses. (2022). <https://arxiv.org/abs/2210.10634> arXiv: 2210.10634 [cs. IR].